

CGA

Sem 4

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===== SOURCE_FILE: sem 4/CGA/PYQs/All units MCQs.pdf =====

CATEGORY: PYQs

STATUS: ok

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Unit-1: Introduction to Computer Graphics

Computer Graphics is primarily concerned with

- A) Data compression
- B) Image creation and manipulation
- C) Networking
- D) Database design

Answer: B

Raster scan displays draw images using

- A) Random vectors
- B) Beam tracing
- C) Pixel-by-pixel scanning
- D) Plotters

Answer: C

Random scan systems are also called

- A) Raster displays
- B) Vector displays
- C) LCD displays
- D) Plasma displays

Answer: B

The smallest addressable unit on a display is

- A) Frame
- B) Pixel
- C) Byte
- D) Vector

Answer: B

Which algorithm is used for drawing a straight line efficiently?

- A) DDA
- B) Flood Fill
- C) Scan-line

D) Ray tracing

Answer: A

Bresenham's algorithm is mainly used for

A) Filling polygons

B) Line drawing

C) Clipping

D) Shading

Answer: B

Midpoint circle algorithm is used to generate

A) Parabolas

B) Ellipses

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C) Circles

D) Polygons

Answer: C

An ellipse has

A) One focus

B) Two foci

C) Three foci

D) No focus

Answer: B

Pixel addressing refers to

A) Assigning memory to CPU

B) Identifying pixel location in frame buffer

C) Drawing curves

D) Clipping

Answer: B

Scan-line fill algorithm is used for

A) Drawing lines

B) Filling polygons

C) Clipping

D) Reflection

Answer: B

Flood-fill algorithm requires

A) Seed point

B) Matrix

C) Vector

D) Projection

Answer: A

Boundary-fill algorithm fills area until

A) Screen ends

B) Boundary color is found

C) Memory ends

D) CPU stops

Answer: B

Hard copy devices include

A) Monitor

B) Plotter

C) Scanner

D) GPU

Answer: B

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Input device used for graphics is

A) Mouse

B) Printer

C) Speaker

D) CPU

Answer: A

Frame buffer stores

A) Program code

B) Pixel intensity values

C) Audio data

D) Network data

Answer: B

Resolution of a monitor is measured in

A) Hertz

B) Inches

C) Pixels

D) Bytes

Answer: C

Aspect ratio is

A) Width/Height

B) Height/Width

C) Pixels/Memory

D) CPU/Memory

Answer: A

Graphics workstation is mainly used for

A) Gaming only

B) Word processing

C) High-end graphics processing

D) Browsing

Answer: C

Parallel curve algorithms generate

A) Straight lines

B) Offset curves

C) Circles only

D) Random points

Answer: B

Refresh rate of display is measured in

A) Pixels

B) Hertz

C) Bytes

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D) Frames

Answer: B

Unit-2: 2D Transformations (20 MCQs)

Basic 2D transformations include

A) Translation

B) Rotation

C) Scaling

D) All of the above

Answer: D

Translation matrix in 2D is of order

A) 2×2

B) 3×3

C) 4×4

D) 2×3

Answer: B

Homogeneous coordinates add

A) One extra dimension

- B) Two extra dimensions
- C) No dimension
- D) Time

Answer: A

Composite transformation means

- A) Single transformation
- B) Multiple transformations combined
- C) Reflection only
- D) Scaling only

Answer: B

Shearing changes

- A) Size
- B) Orientation
- C) Shape
- D) Color

Answer: C

Reflection about x-axis changes

- A) x coordinate
- B) y coordinate sign
- C) Both
- D) None

Answer: B

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Rotation matrix uses

- A) $\tan\theta$
- B) $\sin\theta$ and $\cos\theta$
- C) $\log\theta$
- D) $e\theta$

Answer: B

Viewing pipeline transforms

- A) Model to screen coordinates
- B) Screen to model
- C) CPU to GPU
- D) File to memory

Answer: A

Clipping removes

- A) Visible parts

- B) Outside portions
- C) Color
- D) Resolution

Answer: B

Cohen-Sutherland algorithm is for

- A) Circle drawing
- B) Line clipping
- C) Polygon filling
- D) Scaling

Answer: B

Liang-Barsky algorithm is more efficient because

- A) Uses parametric equations
- B) Uses recursion
- C) Uses matrices
- D) Uses shading

Answer: A

Nicholl-Lee-Nicholl algorithm is used for

- A) Polygon clipping
- B) Line clipping
- C) Filling
- D) Projection

Answer: B

Window is

- A) Entire screen
- B) Selected world region

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- C) Pixel
- D) Memory

Answer: B

Viewport is

- A) World coordinates
- B) Device coordinate region
- C) CPU
- D) GPU

Answer: B

Point clipping tests whether

- A) Inside window

- B) Outside screen
- C) Pixel is colored
- D) Matrix valid

Answer: A

Reflection matrix determinant is

- A) 0
- B) 1
- C) -1
- D) 2

Answer: C

Order of transformations matters in

- A) Translation only
- B) Scaling only
- C) Composite transformations
- D) None

Answer: C

Affine transformations preserve

- A) Parallelism
- B) Length
- C) Angles
- D) Color

Answer: A

Rotation about origin keeps

- A) Distance constant
- B) Area zero
- C) Color same
- D) Pixel same

Answer: A

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Clipping of concave polygon requires

- A) Splitting
- B) Scaling
- C) Shearing
- D) Rotation

Answer: A

Unit-3: 3D Transformations

3D translation matrix is

- A) 3×3
- B) 4×4
- C) 2×2
- D) 5×5

Answer: B

3D rotation can be about

- A) X-axis
- B) Y-axis
- C) Z-axis
- D) All

Answer: D

Perspective projection gives

- A) Parallel lines remain parallel
- B) Realistic depth effect
- C) No depth
- D) Same size

Answer: B

Parallel projection maintains

- A) Convergence
- B) Parallelism
- C) Vanishing points
- D) Curves

Answer: B

View volume defines

- A) Visible region
- B) CPU memory
- C) Matrix
- D) Pixel

Answer: A

Scaling in 3D affects

- A) Size

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- B) Color
- C) Texture
- D) Memory

Answer: A

Shearing in 3D distorts

- A) Shape
- B) Color
- C) Resolution
- D) Brightness

Answer: A

Homogeneous coordinate in 3D is

- A) (x,y)
- B) (x,y,z)
- C) (x,y,z,w)
- D) (x,z)

Answer: C

Projection plane is

- A) Imaginary plane
- B) CPU
- C) RAM
- D) Pixel

Answer: A

Perspective divide involves dividing by

- A) x
- B) y
- C) z
- D) w

Answer: D

Clipping in 3D removes

- A) Hidden parts
- B) Outside view volume
- C) Color
- D) Texture

Answer: B

Viewing transformation converts

- A) World to eye coordinates
- B) Eye to world
- C) Screen to CPU
- D) None

Answer: A

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Reflection in 3D can be about

- A) Plane
- B) Line
- C) Point
- D) All

Answer: A

Matrix multiplication order is

- A) Commutative
- B) Not commutative
- C) Same always
- D) None

Answer: B

Orthographic projection is a

- A) Perspective
- B) Parallel projection
- C) Scaling
- D) Rotation

Answer: B

Vanishing point occurs in

- A) Parallel projection
- B) Perspective projection
- C) Scaling
- D) Translation

Answer: B

3D pipeline includes

- A) Modeling
- B) Viewing
- C) Projection
- D) All

Answer: D

Rotation preserves

- A) Distance
- B) Shape
- C) Angles
- D) All

Answer: D

View reference point defines

- A) Eye position
- B) Pixel

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C) Matrix

D) CPU

Answer: A

General projection transforms

A) 3D to 2D

B) 2D to 3D

C) 1D to 2D

D) None

Answer: A

Unit-4: Visible Surface Detection

Back-face detection removes

A) Front faces

B) Hidden faces

C) Pixels

D) Colors

Answer: B

Z-buffer method stores

A) Color

B) Depth

C) Texture

D) Matrix

Answer: B

Z-buffer requires memory equal to

A) Number of pixels

B) Number of polygons

C) CPU size

D) None

Answer: A

Depth sorting is also called

A) Painter's algorithm

B) Flood fill

C) DDA

D) Clipping

Answer: A

BSP tree stands for

- A) Binary Space Partitioning
- B) Basic Scan Process
- C) Buffer Space Partition
- D) None

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Answer: A

Ray casting determines visibility by

- A) Sorting
- B) Shooting rays
- C) Filling
- D) Clipping

Answer: B

Scan-line method works

- A) Line by line
- B) Pixel by pixel
- C) Frame by frame
- D) Randomly

Answer: A

Area subdivision divides

- A) Objects
- B) Screen area
- C) CPU
- D) Memory

Answer: B

Illumination model calculates

- A) Light intensity
- B) Pixel location
- C) Memory
- D) CPU speed

Answer: A

Ambient light is

- A) Direct light
- B) Background light
- C) Reflected light
- D) None

Answer: B

Diffuse reflection depends on

- A) Viewer
- B) Light angle
- C) CPU
- D) Memory

Answer: B

Specular reflection creates

- A) Matte surface

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- B) Highlight
- C) Shadow
- D) Texture

Answer: B

Ray tracing handles

- A) Reflection
- B) Refraction
- C) Shadows
- D) All

Answer: D

Wireframe model shows

- A) Surfaces
- B) Edges only
- C) Colors
- D) Textures

Answer: B

Polygon rendering fills

- A) Edges
- B) Surfaces
- C) Memory
- D) CPU

Answer: B

Halftoning simulates

- A) Depth
- B) Color variation
- C) Continuous tone
- D) Texture

Answer: C

Dithering reduces

- A) Color banding
- B) Memory
- C) CPU
- D) Size

Answer: A

Hidden surface removal improves

- A) Speed
- B) Realism
- C) Memory
- D) None

Answer: B

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Ray tracing is computationally

- A) Cheap
- B) Expensive
- C) Simple
- D) None

Answer: B

Z-buffer compares

- A) Color
- B) Depth values
- C) Intensity
- D) Texture

Answer: B

Unit-5: Computer Animation & Color Models

Animation is based on

- A) Persistence of vision
- B) Reflection
- C) Clipping
- D) Scaling

Answer: A

Key framing defines

- A) Intermediate frames
- B) Important poses
- C) Pixels
- D) Colors

Answer: B

Tweening generates

- A) Key frames
- B) Intermediate frames
- C) Pixels
- D) Matrices

Answer: B

RGB is a

- A) Subtractive model
- B) Additive model
- C) Binary model
- D) Vector model

Answer: B

CMY model is

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- A) Additive
- B) Subtractive
- C) Binary
- D) None

Answer: B

HSV stands for

- A) Hue, Saturation, Value
- B) High, Speed, Voltage
- C) Hue, Shade, Vector
- D) None

Answer: A

Color depth defines

- A) Brightness
- B) Bits per pixel
- C) Pixels
- D) Resolution

Answer: B

JPEG uses

- A) Lossless compression
- B) Lossy compression
- C) No compression
- D) Binary

Answer: B

PNG is generally

- A) Lossy
- B) Lossless
- C) No format
- D) Video

Answer: B

Contrast stretching improves

- A) Brightness range
- B) Memory
- C) CPU
- D) Size

Answer: A

Digital image consists of

- A) Pixels
- B) Vectors
- C) Lines
- D) Curves

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Answer: A

Matrix in graphics is used for

- A) Transformations
- B) Coloring
- C) Filling
- D) Clipping

Answer: A

Deformation changes

- A) Shape
- B) Color
- C) Memory
- D) CPU

Answer: A

Open-source image software example

- A) Photoshop
- B) GIMP
- C) MS Paint
- D) CorelDraw

Answer: B

24-bit color supports

- A) 16 colors
- B) 256 colors
- C) 16 million colors
- D) 2 colors

Answer: C

Vector graphics are resolution

- A) Dependent
- B) Independent
- C) Fixed
- D) None

Answer: B

Frame rate measured in

- A) Hz
- B) FPS
- C) Pixels
- D) Bytes

Answer: B

Motion specification controls

- A) Speed & path

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- B) Color
- C) Memory
- D) CPU

Answer: A

Additive color mixing uses

- A) RGB
- B) CMY
- C) HSI
- D) None

Answer: A

Compression reduces

- A) File size
- B) Resolution
- C) CPU
- D) Color

Answer: A

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CATEGORY: PYQs

STATUS: ok

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SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
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Academic Year 2025 - 2026 (EVEN)
21CSE255T COMPUTER GRAPHICS AND ANIMATION
UNIT WISE QUESTIONS BANK-SCENARIO BASED

UNIT 1

1. List three use cases that explains DDA line drawing algorithm
2. List out the output primitives. Identify the curve attributes and its type with the help of neat diagram
3. You are developing a graphics application that requires drawing a line efficiently on a pixel grid. To achieve this, you decided to use Bresenham's Line Drawing Algorithm, which minimizes computational complexity while ensuring accurate pixel plotting. The task is to draw a line from Starting point (2, 1) and ending point (8,5) on a pixel grid. Finally, visualize the computed pixel coordinates on a grid to verify the accuracy of the line drawing.
4. Developer needs to design a graphics and animation model. Can you suggest any five applications of computer Graphics briefly

5. What is scan line conversion? Compare and contrast about Random scan and Raster scan with an neat diagram
6. Use Midpoint circle algorithm calculate the pixel that will be with an origin center (0,8) of a circle with radius of 8
7. Illustrate with neat diagram about the following Video display devices:
 - (a) Beam Penetration method
 - (b) Shadow mask method
8. Derive the decision parameter for "Mid-point circle Algorithm"
9. List three use cases that explains Bresenham's line drawing algorithm
10. List out the output primitives. Identify the Line attributes and its type with the help of neat diagram
11. You are working on a graphics program that needs to draw a line on a pixel grid as efficiently as possible. You choose to employ DDA Line Drawing Algorithm, which ensures precise pixel plotting while reducing computing complexity, to do this. Drawing a line on a pixel grid from the starting position (4,2) and ending point (10,7) is the job. Lastly, check the correctness of the line drawing by seeing the calculated pixel coordinates on a grid.
12. A developer needs to design a graphics and animation model. Can you suggest any five applications of computer Graphics briefly.

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13. Explain in detail about CRT monitors and color CRT monitors with an neat diagram
14. Write brief about any two open-source software related to computer graphics and animation and illustrate with an example
15. In a digital clock application, the designer needs to draw a circular clock face with a diameter of 18 units on a pixel grid. To ensure smooth and efficient rendering, the midpoint circle drawing algorithm is used to determine the precise pixel position that form the circle. Given a centre point, apply the algorithm to calculate the decision parameter at each step, determine the pixels for the first quadrant, use symmetry to complete the circle
16. Derive the decision parameter for "Bresenham's Line drawing Algorithm" and how it differs from the DDA Line Drawing Algorithm
17. Explain in detail about the 2D transformation and other transformations with an Example
18. Difference between DDA Line Drawing Algorithm and Bresenham's Line Drawing Algorithm

19. List out the output primitives. Identify the Area fill attributes, Pattern fill attributes, text attribute and marker attribute.
20. Drawing a line on a pixel grid effectively is a requirement of the graphics program you are creating. To do this, you used the Bresenham's Line Drawing Algorithm, which reduces computing complexity while maintaining precise pixel plotting. On a pixel grid, draw a line from the Starting point (2,2) and ending point (9,6). Finally, display the obtained pixel coordinates on a grid to ensure the line drawing is accurate.
21. Illustrate with neat diagram about the following Video display devices:
- i. Direct view storage tube
 - ii. Colour CRT
22. List the types of scans and its operating characteristics for the display technologies with a neat sketch
23. Use Midpoint circle algorithm calculate the pixel that will be with an origin center (0,10) of a circle with radius of 10
24. You are programming a low -power microcontroller to draw thousands of lines per second. Basic Line with Gentle Positive Slope ($0 < m < 1$) and developing a raster graphics system using the decision parameter for "Bresenham's line drawing Algorithm"
25. You are designing software for a low -power embedded display. The system must draw thousands of lines per second and derive the decision parameter for DDA Line drawing Algorithm with the slope ($m > 1$) and ($m = 1$)
26. Use Midpoint circle algorithm calculate the pixel that will be with an origin center (0,7) of a circle with radius of 7
27. Explain briefly about input and output devices
28. Illustrate with neat diagram about the following Video display devices:
- iii. Shadow mask method
 - iv. Flat panel display
29. You are developing a graphics application that requires drawing a line efficiently on a pixel grid. To achieve this, you decided to use DDA Line Drawing Algorithm, which minimizes computational complexity while ensuring accurate pixel plotting. The task is to draw a line from Starting point (1, 7) and ending point (11,17) on a pixel grid. Finally, visualize the computed pixel coordinates on a grid to verify the accuracy of the line drawing.

UNIT 2

1. Demonstrate in detail the 2D transformation and how two sequential reflections around the coordinate axes are equal to a single rotation of translation transformation using an example in matrix representation

2. Discuss about matrix representation and give the composite transformation for Scaling

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3. Show the two successive translation about either of the coordinate axis is equivalent to a

single shearing about the coordinate origin

4. Illustrate with given example A(4,9) B(7,13) and C(10,16) by 3 units in X direction and 4 in

Y directions, which available in two dimensional coordinate transformations for Translation and scaling

5. Discuss about matrix representation and homogeneous coordinates and give the composite transformation for translation and rotation

6. Explain in detail about the 2D transformation and three successive reflections about the coordinate axes is equal to single rotation of other transformation with an Example

7. A student is working on a computer graphics program to display only the visible part of objects inside a screen. A rectangular viewing window is defined by points: A (20,20), B (60,20), C (60,40), D (20,40). A line is drawn between two points: P (40,80) and Q (120,30), but part of the line lies outside the window. Use the Cohen-Sutherland line clipping algorithm

to find the visible portion of the line inside the window. Determine the final clipped line coordinates.

8. Given a clipping window A (10, 10), B (70, 10), C (70, 50), D (10, 50). Using Cohen Sutherland and algorithm find the visible portion of line segment joining the points X (30, 90) and Y (100, 20). In a game rendering system, a polygon representing an object is partially

outside the screen. For a polygon and clipping window shown in figure below give the list of vertices after each boundary clipping. Clip the polygon against the window using Sutherland-Hodgman algorithm and find the final visible polygon.

9. Imagine you are developing a navigation display system for a drone. The drone's camera only shows a rectangular viewing window on the screen. The visible screen area (window) is bounded by: Lower -left corner: (50, 10), Upper -right corner: (80, 40) While tracking objects, the drone detects a path segment (a straight line) between two points: P1 = (70, 20)

→ starting point of the path, P2 = (100, 10) → ending point of the path, However, parts of this path may lie outside the visible screen. Using the Cohen -Sutherland line clipping

algorithm, determine: Whether the line segment is completely visible, partially visible, or invisible within the window. If it is partially visible, calculate the clipped line segment coordinates that will appear on the screen. Show the region codes for both endpoints and explain each step of the clipping process.

10. You are developing a 2D animation editor where users draw lines that extend beyond the screen

boundary. Explain how the Cohen-Sutherland line clipping algorithm can be applied to clip the

lines within the viewport with example.

11. A developer is working on a graphics system where multiple lines must be clipped efficiently

within a viewing window. To improve the performance, the developer chooses the Nicholl-Lee-Nicholl (NLN) line clipping algorithm instead of traditional methods. Explain the Nicholl-Lee-

Nicholl line clipping algorithm in simple terms. Describe how it improves efficiency compared to

other clipping algorithms with a neat diagram.

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12. A student is using a graphics software to edit shapes: The student draws a triangle with points

A (2,2), B (4,6), C (6,2) and applies a mirror effect. Find the new coordinates after reflection about:

X-axis, Y-axis, Origin

13. The student draws a rectangle A (0,0), B (0,4), C (3,4), D (3,0) and applies a shear effect. Find the

new coordinates after shearing with: $S_x=2$ $S_x = 2$ $S_x=2$, $S_y=2$, $S_y = 2$, $S_y=2$

14. You are working as a graphics developer for a 2D animation software used in designing maps and

game environments. In one scene, a hexagonal object (shaded region) partially overlaps with a

rectangular viewport (display window), as shown in the figure.

15. Explain in detail about the 2D transformation and two successive reflection about the coordinate axes is equal to single rotation of other transformation with an Example

16. Illustrate with given example A(3,8) B(6,12) and C(9,15) by 4 units in X direction and 5 in Y

directions, which available in two dimensional coordinate transformations for Translation

and

scaling

17. Discuss about matrix representation and homogeneous coordinates and give the composite transformation for translation and rotation

UNIT 3

1. A developer is creating a 3D animation where objects need to move, rotate, and resize smoothly in a scene. List the basic 3D transformations and explain. Discuss how these transformations are represented using matrix operations.
2. A virtual camera system is implemented in an animation software. Explain the 3D viewing pipeline (modeling, viewing, projection, viewport).
3. Evaluate parallel vs perspective projection in terms of realism and computational efficiency.
4. You are developing a 3D animation scene where an object moves, rotates, and scales simultaneously. Derive the composite transformation matrix.

5. A game development company is designing a 3D animation engine where objects must be moved, resized, and rotated dynamically before rendering. A car model is initially positioned at (1, 2, 3) (1, 2, 3) (1, 2, 3). It is required to: Move forward by 10 units along the X-axis Rotate 45° about the Y-axis Scale uniformly by a factor of 1.5. Formulate the composite transformation matrix. Determine the order of transformations and its impact on the final output.

6. A point $P(2, 3, 4)$ is subjected to the following sequence of transformations in a 3D

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animation system: Translate the point by (3, -2, 5), Rotate the point about the Z -axis by 90° and Scale the object by factors $S_x = 2$, $S_y = 1$, $S_z = 3$

7. A company is developing a 3D flight simulation system where real-world terrain, aircraft, and environmental objects must be rendered accurately onto a 2D display screen. Describe the complete 3D viewing pipeline involved in transforming the 3D scene into a 2D image. Compare parallel projection and perspective projection, and justify which is more suitable for this simulation.

8. A Design Engineer 3D graphics application where multiple objects overlap on the screen. To display only the visible parts correctly, different visible surface detection methods are considered. Classify the visible surface detection methods used in computer graphics and explain each category with examples.

Given a 3D triangle with points $A(0, 0, 0)$, $B(3, 3, 4)$ and $C(3, 3, 5)$. Apply shear parameter 2 on X axis, 2 on Y axis and 2 on Z axis and find out the new coordinates of the object.

Given a 3D triangle with coordinate points $A(0, 0, 0)$, $B(3, 3, 4)$ and $C(3, 3, 5)$. Apply the

reflection on the XY, YZ and XZ plane and find out the new coordinates of the object

9. A 3D object is placed above a mirror lying on the ground (XY -plane). You want to find the reflected image. A building model is pushed sideways due to wind, causing the top to shift while the base remains fixed. In computer graphics engines (like games), all transformations (including translation) are handled uniformly.

An engineer is drawing a machine part where the front face must be exact, and depth should also be visible without using perspective and a cube drawn using oblique projection looks too stretched in depth.

10. You stand on a railway track and observe that the rails seem to meet at a distance. Why do

parallel lines appear to meet in perspective projection? Find and state which projection was used for the given scenario "An artist draws a cube where all depth lines converge to one point"

11. You are a graphics programmer working for a game development company. Your team is designing a realistic 3D environment for a new video game that includes multiple objects, complex geometries, and reflective surfaces like mirrors and water

12. While rendering the scene, you notice that some objects incorrectly appear in front of others, even when they should be hidden. To fix this, you need to implement appropriate visible surface detection methods.

13. The game also requires highly realistic reflections on surfaces like glass walls and shiny

floors using ray tracing

14. Clip the line L1 with the endpoints (7, 40) and (50, 115) using the Nicholl Lee Nicholl line

clipping algorithm. (Refer the following figure).

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15. Illustrate about Composite 3D Transformation? You are designing a 3D drone simulation system where a drone manipulates objects in space (x, y, z). The drone performs:

- moving objects (translation)
- rotating objects for alignment
- resizing objects (scaling)

To improve efficiency, you must compute composite transformation matrices

16. A rectangular parallelepiped is given having length on x-axis, y-axis and z-axis as 3, -2

and 1 respectively. Perform a rotation by an angle -90° about x-axis and an angle 90° about y-axis. A cube is placed in space, and a light source directly above casts a shadow on the ground (XY-plane). Find the projection matrix and the projected coordinates of point P(3,4,5). State Shadow on Ground (Orthographic Projection) with neat sketch.

17. A mechanical engineer is designing a machine component using a CAD system. The engineer wants all dimensions (length, width, height) to remain accurate without any distortion, even if objects are far away apply Orthographic Projection on it.

UNIT 4

1. A computer graphics engineer is developing a rendering system where some scenes require processing based on object geometry, while others rely on pixel -level visibility calculations. Categorize the following visible surface detection methods into object-space and image-space techniques: Scan-line, BSP tree, Area Subdivision, Octree.

2. A production studio wants to improve realism in animated scenes. Compare surface rendering methods and ray tracing techniques. Justify which method is more appropriate for high-quality animation output.

3. A developer is designing a real -time computer graphics application for an interactive system. During development, the system faces performance inefficiencies due to rendering surfaces that are not visible to the viewer. Explain how Back-Face Detection can be used to eliminate surfaces not visible to the viewer. Describe the Scan-Line Method for visible surface detection and illustrate its working with a neat diagram.

4. A designer is developing a computer graphics scene in which the objects appear unrealistic due to improper lighting. Some surfaces look dull, and reflective objects fail to display highlights correctly. What are the basic illumination models and brief them. Write the intensity equations for each model.

5. A large -scale 3D animation scene with complex geometry suffers from performance issues. Discuss how BSP Tree and Octree methods help in visible surface detection.

6. An animator observes banding effects in smooth color gradients of a rendered scene. Explain how Halftoning and Dithering techniques reduce such artifacts.

7. You are implementing a visibility detection system for curved surfaces in a 3D animation. Explain the working of the Ray Casting method. Evaluate its advantages compared to traditional visibility detection techniques.

8. List the contents of vertex table, edge table and surface table for the following polygon shown in figure

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9. You are creating a realistic animated forest scene with sunlight effects. Explain the role of illumination models (ambient, diffuse, specular reflection). Evaluate how lighting

improves depth perception in animation

10. Compare and contrast the given scenario: The scene contains many overlapping polygons, and it is difficult to determine which surfaces are visible. In this case, the engineer uses the Area Subdivision Method, where the screen is recursively divided into smaller regions until visibility can be easily decided. After rendering, the image shows jagged edges (aliasing effect) and lacks smoothness. Here, the engineer applies Antialiased Ray Tracing, where multiple rays are traced per pixel and their colors are averaged to produce smooth and realistic edges.

11. Describe the Back -Face Detection Method. This method is very useful in the visible - surface detection process, why? What else should be handled after Back -Face Detection, in order to conclude the set of all visible-surfaces?

12. List the contents of vertex table, edge table and surface table for the following polygon shown in figure.

13. You are drawing multiple transparent glass sheets stacked in space. Which rendering method ensures correct visibility in an illumination model calculates how light interacts with polygon surfaces to determine their color and brightness.

i. A low-poly game renders objects where each face has a single color.

ii. A 3D object looks smoother, but highlights are slightly dull.

A shiny object (like a metal ball) shows bright highlights that move with the viewer.

14. An animator notices visual artifacts (banding) in gradient regions of a rendered scene using phong and warn model. Explain how half toning and dithering techniques improve image quality. Analyze their importance in digital animation output.

UNIT 5

1. State the various color models and explain them.

2. Design and develop a Two -Dimensional cartesian Reference frames with real -time

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example using right handed system

3. Compare and contrast between the RGB and CMY color models.
4. Summarize about CIE color model. What are its advantages?
5. Given a colour represented in RGB colour space as $R = 0.2$, $G = 0.6$, $B = 0.3$, Convert from RGB to CMYK and YIQ colour models. Explain CMYK and YIQ coloring models with their significances.
6. Write in details about numerical methods in computer graphics. consider own dataset for "Fitting Curves"
7. Consider a full tone digital image size 1024×768 with 8-bit representations, answer the following questions:
 - i. How will you capture digital images?
 - ii. Which tool would you prefer to use when you want to add noise in given image in adobe Photoshop?
 - iii. What kind of tools would you prefer to use when you want to select the part of image or whole image in adobe Photoshop?
 - iv. How will You change color modes of the image in Adobe photoshop?
 - v. How will you apply lighting effects to a digital image in Adobe Photoshop?